

Applied SVAR Identification

Day 3: Identification II — Long-Run Restrictions

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Course Roadmap

Day	Topic
1	VAR Foundations: Motivation, Estimation, Specification, Identification & Tools
2	Identification I: Short-Run Restrictions
3	Identification II: Long-run Restrictions
4	Identification III: Sign, Magnitude, and Narrative Restrictions
5	Identification IV: Instruments

Learning Objectives

By the end of today, you should be able to:

1. Explain what sign, magnitude and narrative identification are and how they each solve the identification problem.
2. Understand the limitations of sign restrictions for identification.
3. Implement sign, magnitude and narrative identification in MATLAB using the ACB Toolbox.
4. Apply sign, magnitude and narrative identification to replicate Uhlig (2005), Kilian and Murphy (2012), and Antolín-Díaz and Rubio-Ramírez (2018).

Part 1

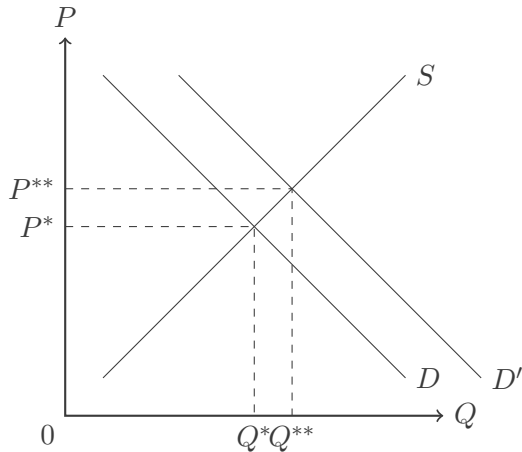
Sign Restrictions

Restricting the sign/direction of the impact of structural shocks

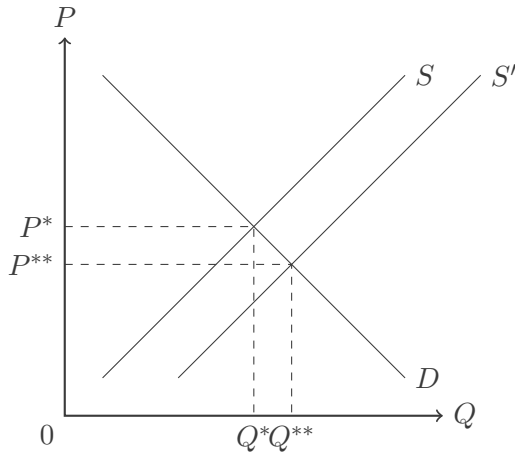
Background: Identification so far

- ▶ Sims (1980) introduced VARs to economists along with short-run recursive restrictions as an identification strategy
- ▶ Cooley and LeRoy (1985) criticized many recursive VARs for being atheoretical
- ▶ Led to the development of non-recursive short-run zero restrictions (e.g., Bernanke (1986), Blanchard and Watson (1986))
- ▶ Also inspired Blanchard and Quah (1989) to use long-run macroeconomic theory to justify long-run zero restrictions
- ▶ **Another Idea:** Use theory to motivate restrictions on the sign (direction) of a variables response to a shock (Faust (1998); Canova and De Nicolò (2002); Uhlig (2005)) – called *sign restrictions*

Example: Market Supply and Demand Model



Positive demand shock $\rightarrow \uparrow P$ and $\uparrow Q$



Positive supply shock $\rightarrow \downarrow P$ and $\uparrow Q$

Example: Market Supply and Demand Model

- ▶ Economic theory suggests the following signs

	$\varepsilon_{D,t}$	$\varepsilon_{S,t}$
p_t	+	+
q_t	+	-

1. Positive demand shock $\Rightarrow \uparrow p_t$ & $\uparrow q_t$
 2. Positive supply shock $\Rightarrow \uparrow p_t$ & $\downarrow q_t$
- ▶ We can therefore place these *sign restrictions* on the impact matrix. How to do this in practice?

Implementing sign restrictions

- **Identification problem:** Use VAR residuals to recover the SVAR shocks, i.e.

$$e_t = B_0^{-1} \varepsilon_t$$

- **Recursive identification:** Since $Cov(e_t) = \Sigma$ is observed, use that to identify the impact matrix via:

$$\Sigma = B_0^{-1} (B_0^{-1})' = PP'$$

where P is the lower cholesky factor of Σ . This makes the structural shocks 'orthogonal', but has zero values so won't satisfy the sign restrictions

- **Idea:** Can we transform this in a way that preserves orthogonality, removes the zeros, and gives the desired sign restrictions? Yes!

Orthonormal/Orthogonal Matrices

- ▶ **Definition:** A matrix Q is an **orthonormal (AKA orthogonal) matrix** if it's square ($n \times n$) and it's columns and rows are **orthonormal vectors**.
- ▶ Orthonormal vectors are also called orthogonal unit vectors (hence the name):
 1. They are mutually orthogonal: $q_i' q_j = 0$ for all $i \neq j$;
 2. They have unit **length**: $\|q_i\| = \sqrt{q_i \cdot q_i} = 1$.
- ▶ Examples of orthogonal matrices:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

- ▶ Important for us because Orthonormal matrices have a property that the transpose is equal to the inverse, i.e. $Q' = Q^{-1}$, implying that $QQ' = I$.

A clever transformation

- We can therefore use the fact that $QQ' = I$ to get

$$e_t = PP' = \underbrace{PQ}_{B_0^{-1}} \underbrace{Q'P'}_{(B_0^{-1})'}$$

- If Q is a full matrix then the matrix PQ will usually be a full matrix too $\rightarrow$ system is no longer recursive!
- Next Question: How can we find such a matrix?

Finding Q

- ▶ Many methods for generating orthogonal matrices (see [Fry and Pagan \(2011\)](#) for a review). The two most popular are:
 1. Givens rotations
 2. Householder transformations
- ▶ [Rubio-Ramírez, Waggoner and Zha \(2010\)](#) show that [Householder transformations](#) (specifically the [QR decomposition](#)) are a more computationally efficient strategy relative to the [Givens rotations](#), and is the main approach used practice. We therefore use that here.

The Rubio-Ramírez, Waggoner and Zha (2010) AKA RWZ algorithm

Perform D replications of the following steps:

1. Compute the lower Cholesky factor of Σ , called P
2. Draw $K \sim N(0, I)$
3. Apply the *QR decomposition* to $K = QR$, where Q is an orthonormal matrix and R is a triangular matrix
4. Check whether PQ satisfies the sign restrictions:
 - 4.1 If yes, set $B_0^{-1} = PQ$
 - 4.2 If no, discard it

This provides a set of solutions to the identification problem $\rightarrow$ SVAR is **set identified**.
Implement in ACB toolbox with: `VARopt.ident = 'sign';`

Uhlig (2005, JME): Overview

Question: What are the effects of a contractionary monetary policy shock on real economic activity?

Motivation: Recursive SVARs find large, persistent output effects, but results hinge on identifying assumptions about variable ordering. Two well-known anomalies suggest misidentification:

1. Prices *rise* following a monetary tightening — called the **price puzzle**
2. Output effects sensitive to whether output is ordered before or after the policy rate

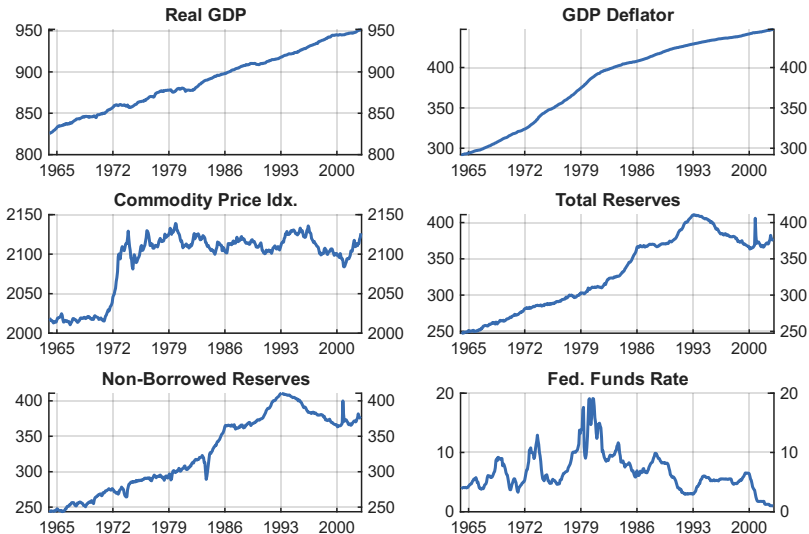
Contribution: Use sign restrictions to address these issues.

Uhlig (2005): The VAR

- ▶ **Monthly SVAR**, 1965:M1–2003:M12; United States. $p = 12$ lags, constant.
- ▶ **Six variables:**
 1. y_t = Real GDP ($\log \times 100$)
 2. π_t = GDP Deflator ($\log \times 100$)
 3. comm_t = Commodity Price Index ($\log \times 100$)
 4. res_t = Total Reserves ($\log \times 100$)
 5. nbres_t = Non-Borrowed Reserves ($\log \times 100$)
 6. ff_t = Federal Funds Rate (%)
- ▶ **Monetary policy shock** identified with sign restrictions imposed for 6 months:

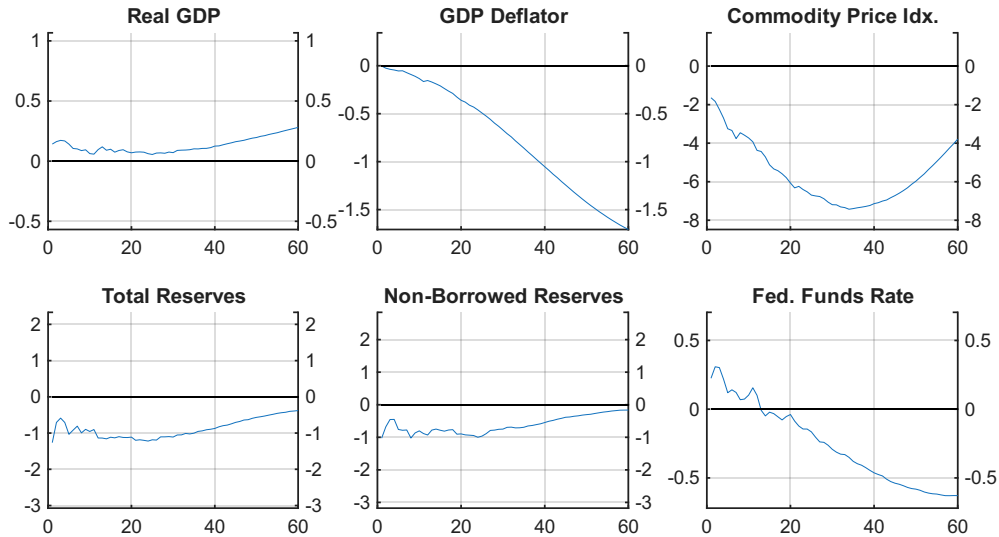
$$[B_0^{-1}]_{:,1} = \begin{bmatrix} 0 \\ - \\ - \\ 0 \\ - \\ + \end{bmatrix} \begin{array}{l} \leftarrow y_t \text{ (unrestricted — agnostic)} \\ \leftarrow \pi_t \\ \leftarrow \text{comm}_t \\ \leftarrow \text{res}_t \text{ (unrestricted)} \\ \leftarrow \text{nbres}_t \\ \leftarrow \text{ff}_t \end{array}$$

Uhlig (2005): Data



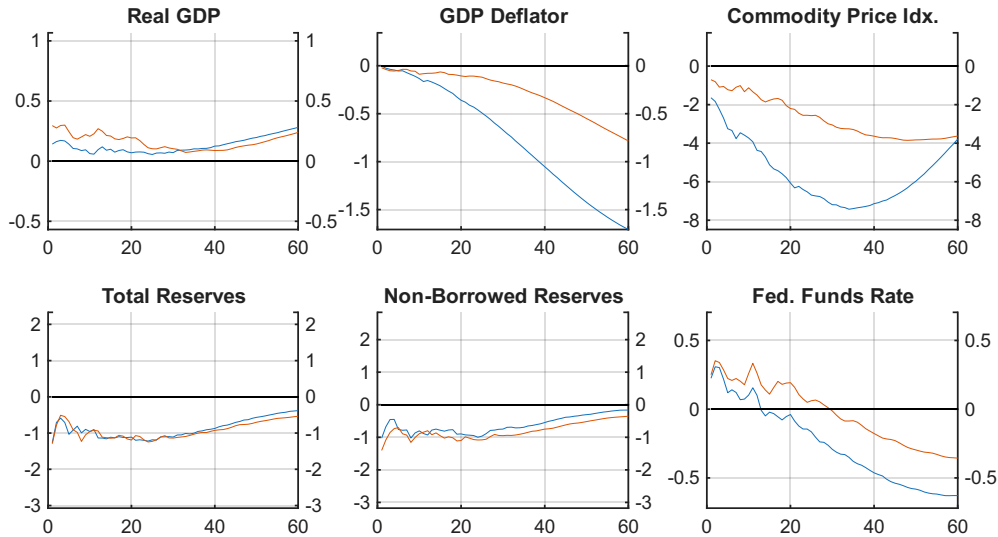
Ex: Describe the evolution of the federal funds rate. When does monetary policy appear most contractionary?

Uhlig (2005): One Accepted Draw



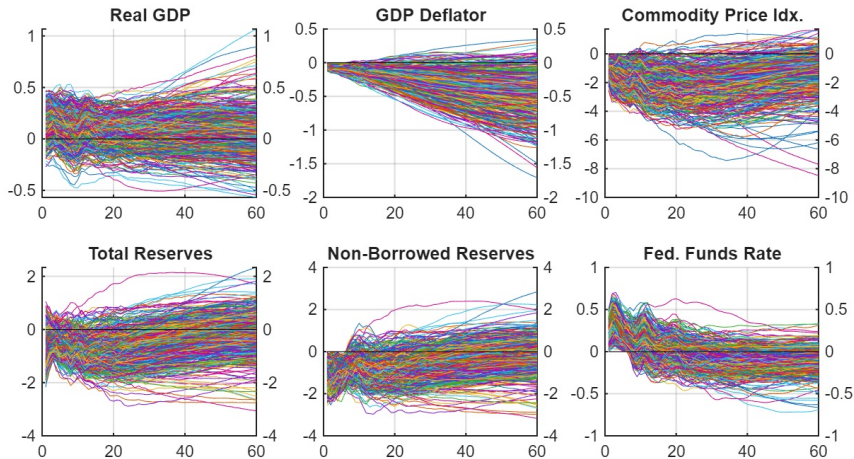
Ex: What is the sign of the output response? Is it consistent with conventional monetary theory?

Uhlig (2005): Two Accepted Draws



Ex: Do both draws agree on the sign of the output response? What does disagreement across draws imply?

Uhlig (2005): All Accepted Draws



Ex: Is the effect of monetary policy on output unambiguously signed? What does the spread of accepted draws tell us about set identification?

Exact identification vs set identification

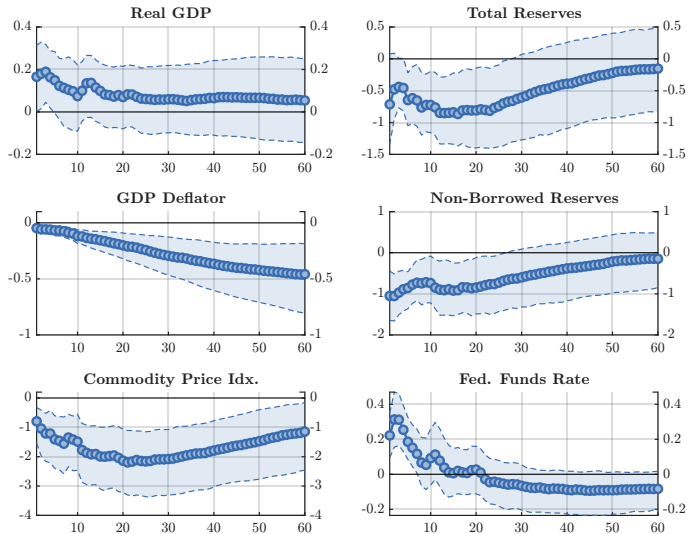
Two types of identification:

- ▶ **Exact identification:** Exactly one solution to the identification problem (e.g., recursive identification) giving unique values for B_0^{-1} – also called **point identification**.
- ▶ **Set Identification:** Many possible solutions to the identification problem (e.g., sign restrictions) giving a set of values for B_0^{-1}

Implications of set identification:

1. A range of possible results IRF/FEVD/HD
2. No single point estimate – which model is best?
3. There is uncertainty from both (1) identification and (2) sample variation (standard errors). This makes things complicated. To accommodate the uncertainty people therefore usually use Bayesian estimation when doing sign restrictions (including ACB Toolbox!).

Uhlig (2005): Median and 1SD bands



Ex: Interpret the results

Some solutions to set identified SVARs

Many possible solutions to getting a representative SVAR from the set of feasible SVARs. Most focus on the IRFs:

- ▶ Uhlig (2005) → uses the median of the set of implied IRFs
- ▶ Fry and Pagan (2011) criticize using the median because it is not representative of an SVAR. Propose choosing the SVAR with impact matrix 'closest to' the median impact matrix (computationally cheap and makes economic sense).
- ▶ Inoue and Kilian (2013; 2019; 2022) argue for the (posterior) mode (requires Bayesian estimation – very computationally intensive)
- ▶ Giacomini and Kitagawa (2021) argue for the mean of 'robust credible region' (requires Bayesian estimation – very complicated)

Notes:

1. ACB toolbox implements both median and Fry–Pagan median, so we use that for IRFs. However, one problem is that median FEVD doesn't sum to 1 → use the mean instead!
2. Lot's of focus on IRFs, but little focus on FEVD or HD — more ideas needed!

Fry and Pagan (2011): Median

Fry and Pagan (2011) get the 'median' SVAR using the following steps:

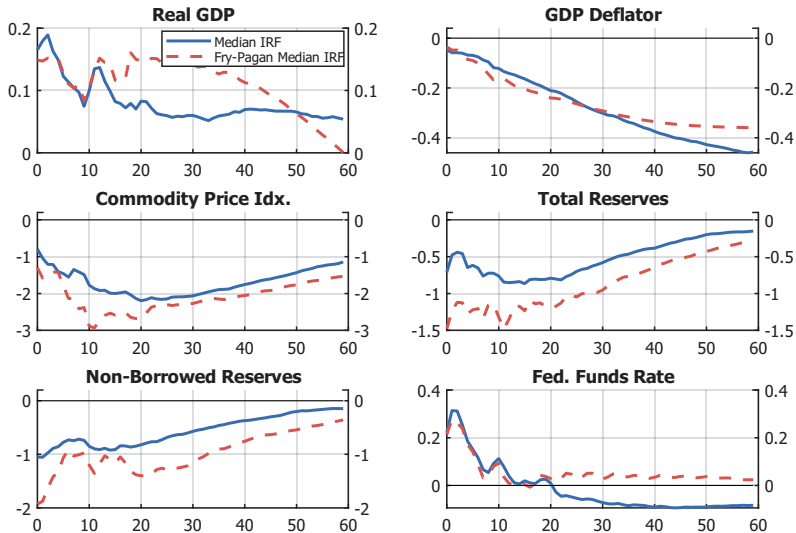
1. **Compute the target:** Take the element-wise median across all accepted draws to form the target matrix B^{med} .
2. **Find the closest draw:** Select the accepted draw B^{fp} that minimises the Frobenius distance to B^{med} .

$$B^{fp} = \arg \min_j \left\| B^{(j)} - B^{med} \right\|_F$$

3. **Report:** Use B^{fp} to construct the point estimates from the SVAR.

ACB Toolbox: Extract the Fry–Pagan impact matrix with 'VAR.Bfp' and implied point estimates for the IRF, FEVD, and HD with 'VAR.IRfp', 'VAR.VDfp', and 'VAR.HDfp', respectively.

Uhlig (2005): Median vs Fry-Pagan Median



Ex: Do the results with Fry-Pagan Median differ?

Sign Restrictions: Pros and Cons

Pros

- ▶ Theory-grounded: restriction signs come directly from economic models
- ▶ Agnostic about magnitudes — no arbitrary zero restrictions or recursive ordering required
- ▶ Flexible: easily combined with narrative or magnitude restrictions to sharpen inference
- ▶ Well-suited when theory is informative about direction but not size of responses

Cons

- ▶ Set-identified: multiple models are admissible — IRF bands reflect model uncertainty, not sampling uncertainty
- ▶ Fry-Pagan problem: element-wise median IRF does not correspond to any valid SVAR
- ▶ Potentially wide identified sets without additional restrictions
- ▶ Sensitive to which signs are imposed — results can be fragile to minor changes in restrictions

Part 2

Narrative Restrictions

Restricting the historical natural of structural parameters

Narrative sign restrictions

- ▶ The main challenge with sign restrictions is that they admit a large set of models, all of which are permissible. This has lead people to think of ways to shrink the set of admissible models and sharpen inference.
- ▶ [Antolín-Díaz and Rubio-Ramírez \(2018\)](#) suggest that one way to shrink the set of admissible sign restrictions is to exploit information from well-documented historical episodes – called **narrative restrictions**

Antolín-Díaz and Rubio-Ramírez (2018): Overview

Question: Build on Uhlig (2005) and ask the same question: What are the effects of a contractionary monetary policy shock on real economic activity?

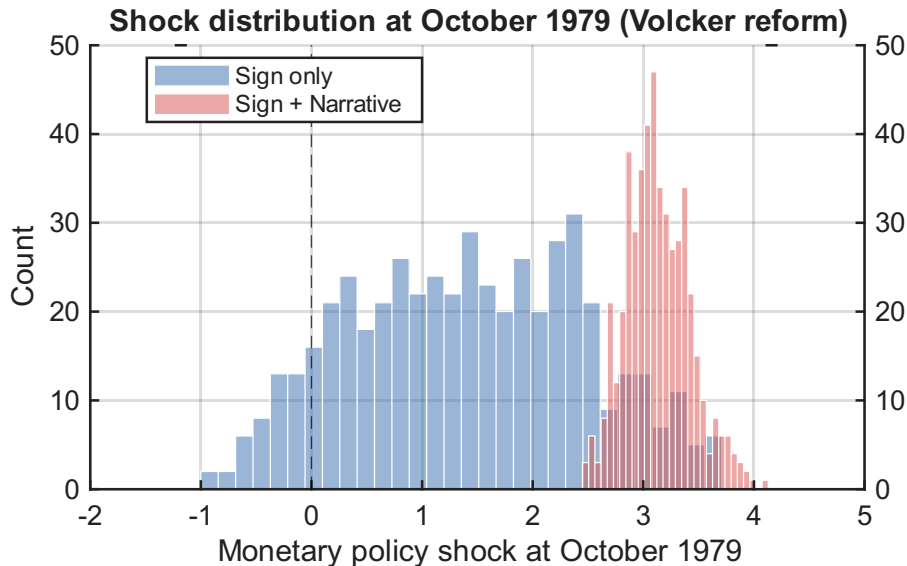
Motivation: Uhlig (2005) presents a large set of admissible models. Can we refine the identification?

Contribution: Use narrative restrictions to thin the set of responses found by Uhlig. Specifically, they argue that the Volcker tightening of October 1979 carries two pieces of important information to help identify monetary policy shocks:

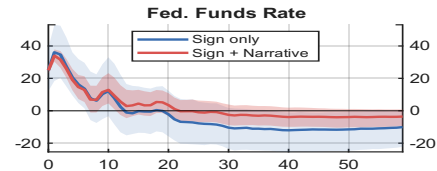
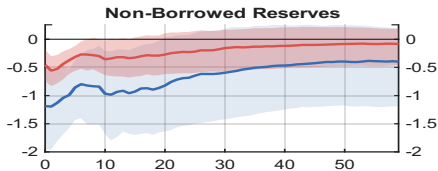
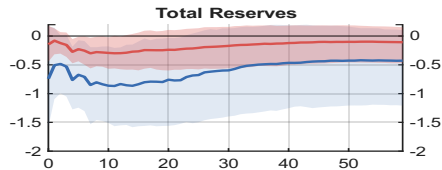
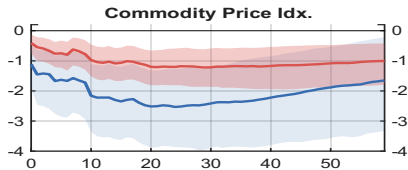
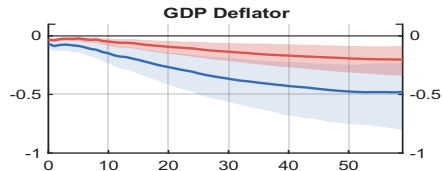
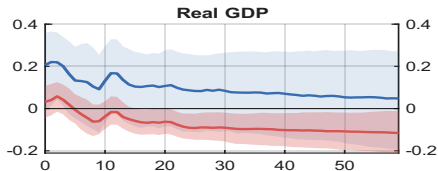
- ▶ The **sign** of the monetary policy shock was a positive;
- ▶ The monetary policy shock was the **dominant driver** of variables during that period.

Implement narrative restrictions in ACB toolbox with: `VARopt.ident = 'narrative';`

Narrative restrictions shrink the identified set of shocks



Narrative restrictions shrink the identified set of IRFs



Part 3

Magnitude Restrictions

Restricting the magnitude/size of structural parameters

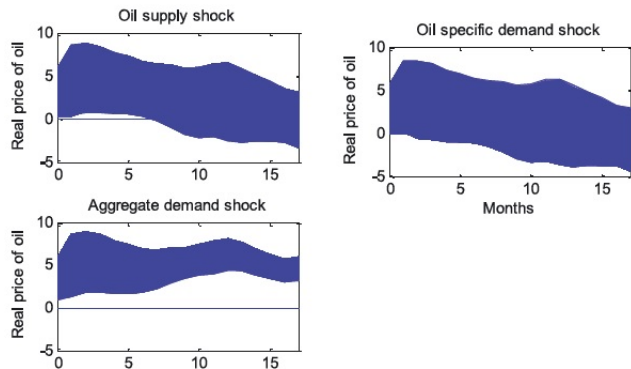
KM (2012): Extending the Kilian (2009) SVAR model

- ▶ The assumption of a short-run oil supply elasticity of zero used in Kilian (2009) is likely not exactly true.
- ▶ Kilian and Murphy (2012) therefore re-do the analysis using sign restrictions:

	Oil supply disruption	Aggregate demand shock	Oil-specific demand shock
Oil production	—	+	+
Real activity	—	+	—
Real price of oil	+	+	+

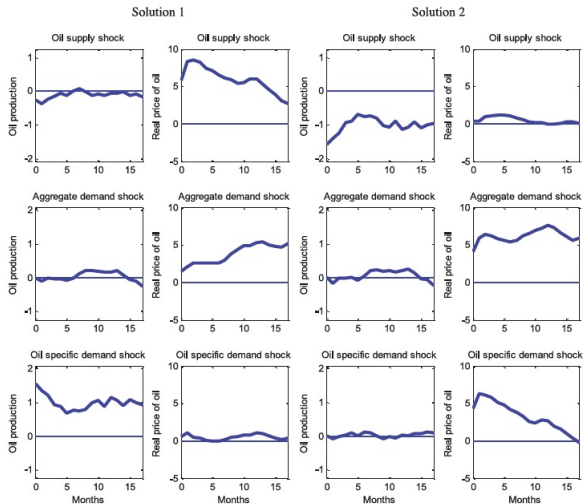
- ▶ To implement the restrictions, they start with 1.5 million random draws of the rotation matrix (i.e. Q). Of these, 30,860 draws satisfy the sign restrictions.

KM (2012): All admissible IRFs



Set of impulse response functions for the real price of crude oil to one-standard-deviation structural shocks, identified by sign restrictions.

KM (2012): Two admissible IRFs

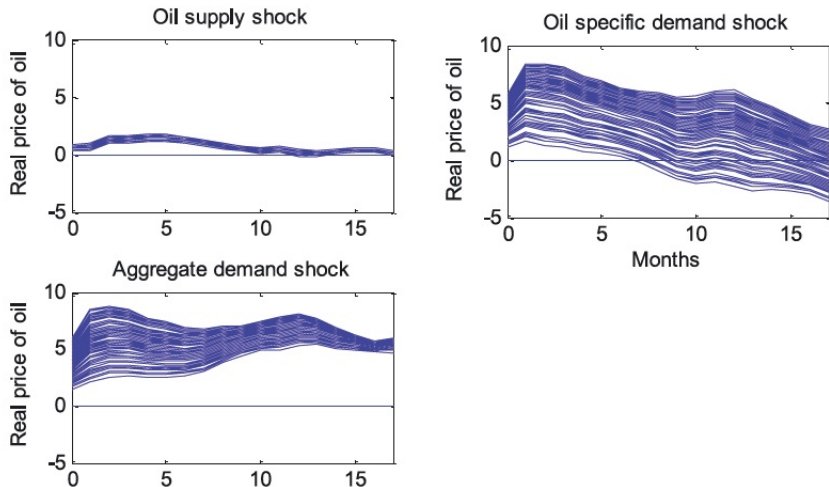


Impulse response functions for the real price of crude oil to one-standard-deviation structural shocks from two candidate SVARs.

Thinning the set of admissible models

- ▶ **Problem:** Both of the SVARs used to generate the IRFs on the previous slide are compatible with the sign restrictions in Table 1, however they both imply very different worlds.
- ▶ **Idea:** Try to thin out the set of possible responses by imposing additional *elasticity restrictions* on the impact matrix.
- ▶ Kilian and Murphy (2012) impose constraints the elasticity of aggregate demand is no smaller than -1.5 and elasticity of supply can be no larger than 0.0258 (see their paper for discussion of this point and demand elasticity restrictions too. Hotly debated...).
- ▶ The supply elasticity means that the following two ratios are less than 0.0258:
 1. Supply response to oil price movements after an aggregate demand shock $\frac{b_{12}}{b_{32}}$.
 2. Supply response to oil price movements after an oil-specific demand shock $\frac{b_{13}}{b_{33}}$.
- ▶ This and the demand elasticity assumption reduces the number of admissible models from 30,860 to 80.

KM (2012): Sign and elasticity restrictions



Set of thinned impulse response functions for the real price of crude oil to one-standard-deviation structural shocks identified by sign and elasticity restrictions.

KM (2012): Comparison with recursive identification (Kilian, 2009)

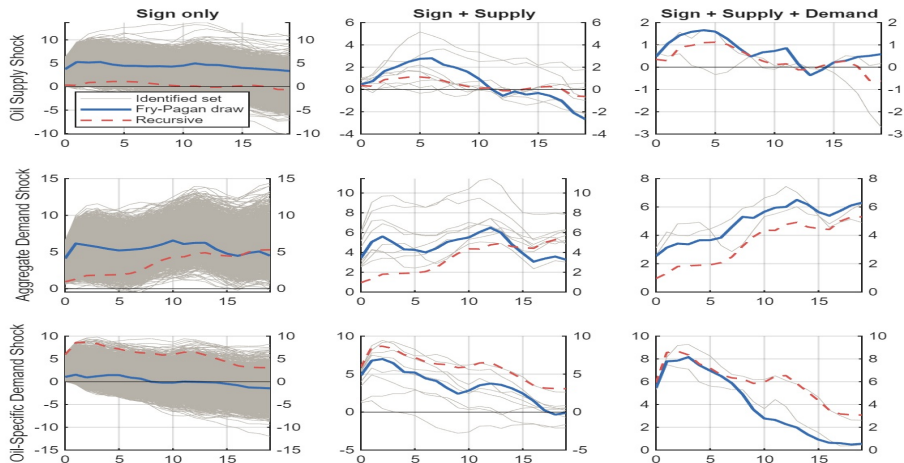
TABLE 4. Alternative estimates of B_0^{-1} .

Recursively Identified Structural Model ^a			Sign Restrictions Combined with All Additional Restrictions		
1.572	0	0	1.567	0.023	0.128
-0.007	4.357	0	0.007	4.252	-0.947
-0.349	0.955	5.924	-0.835	2.211	5.526

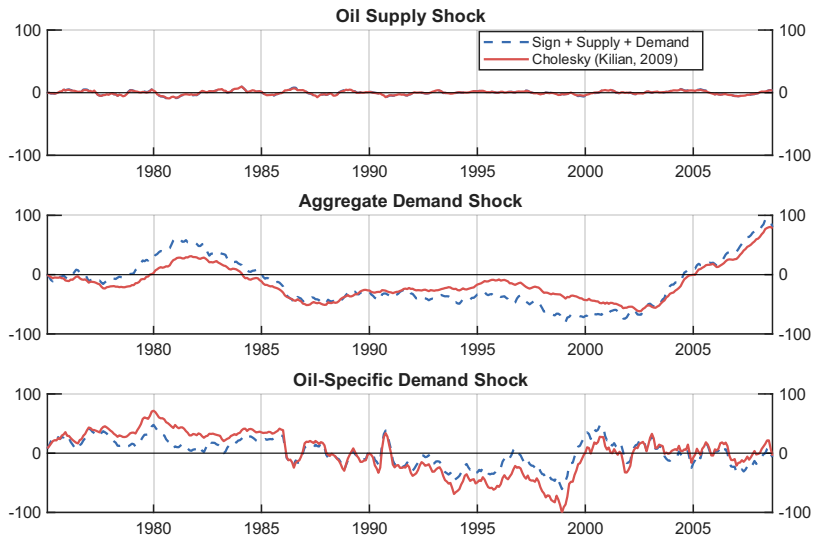
Note: The first column of the identification matrix has signs opposite of the impact responses displayed in the graphs because those graphs depict the effects of an unanticipated oil supply disruption, following the convention adopted in Kilian (2009a).

^a Based on Kilian (2009a).

KM (2012): Identified set of IRFs (replication)



KM (2012): Historical decomposition (replication)



Summary: Sign and Narrative Restrictions

Key ideas from today:

1. **Sign restrictions** identify shocks by the *direction* of impulse responses — no zero restrictions or recursive ordering required
2. **The RWZ algorithm** draws random orthonormal matrices Q , retains those whose implied IRFs satisfy the restrictions, and returns the full set of admissible models
3. **Set identification**: accepted draws span an identified set — IRF bands reflect model uncertainty, not sampling uncertainty.
4. **Best representative model** is debated. [Fry and Pagan \(2011\)](#) median where we select the draw with impact matrix closest to the element-wise median is a defensible choice.
5. **Shrinking the identified set**: two complementary approaches:
 - ▶ [ADRR \(2018\)](#): Narrative restrictions
 - ▶ [KM \(2012\)](#): Elasticity bounds

Preview of Day 5

- ▶ Use high-frequency data to construct **external instruments** to identify structural shocks in **Proxy SVARs** with applications to:
 1. Monetary policy (Gertler and Karadi, 2015)
 2. Oil market shocks (Kánzig, 2021)
- ▶ Use **VARX** models for identification — simpler than Proxy SVAR and permits use of instruments in more complex time-varying parameter VARS (Paul, 2020)